Final Exam

● Graded

Student

Xinrong Sun

Total Points

40.5 / 45 pts

Question 1

Overall Grades **40.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 4.5 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Xinrong Sun	xs156	孙新荣.

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 孙新荣.

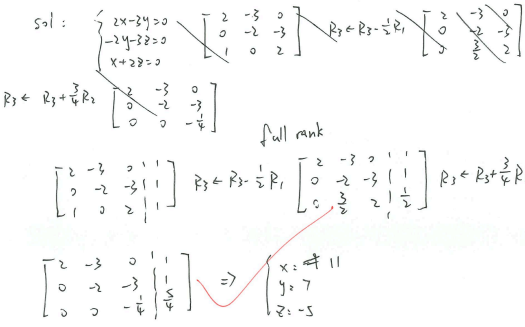
IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	5	0.5	5	40.5

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Problem 1. [5'] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$



Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

sol: (1)

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad R_2 \leftarrow R_2 + \frac{5}{2}R_1 \quad \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 10R_3 = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad R_4 \leftarrow R_4 - \frac{1}{2}R_1 \quad \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$R_4 \leftarrow R_4 - R_3 \quad \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \Rightarrow R_2 \quad \begin{bmatrix} 2 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

a basis of column space: $\#\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

a basis of row space: $\{[2 \quad 4 \quad 2], [-5 \quad 0 \quad 5]\}$

(2)

$$Ax = 0 \quad \text{let } z = t$$
$$\begin{cases} 2x+4y+2z=0 \\ y+z=0 \end{cases} \Rightarrow \begin{cases} x = t \\ y = -t \\ z = t \end{cases}$$

a basis of null space: $\#\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$T(x) = Ax$

so |:

$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

$2\times 3 \quad 3\times 1 \quad 2\times 1$

$A\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$

$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$ $\quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix}$ $\quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$

$\Rightarrow \begin{cases} b-c=1 \\ e-f=1 \end{cases}$ $\quad \begin{cases} a-c=0 \\ d-f=2 \end{cases}$ $\quad \begin{cases} -c=2 \\ -f=2 \end{cases}.$

$\Rightarrow \begin{cases} a=-2 & d=0 \\ b=-1 & e=-1 \\ c=-2 & f=-2 \end{cases}$

Therefore, $A = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$

Problem 4. [5'] Find the Jordan canonical form of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

So:

$$Av = \lambda v$$

$$(A - \lambda I)v = 0$$

$$\det(A - \lambda I) = 0$$

$$\begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)\cdot(-1)^{1+1}\begin{vmatrix} 2-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} + 2\cdot(-1)^{1+2}\begin{vmatrix} 0 & 0 \\ -2 & 1-\lambda \end{vmatrix}$$

$$= (1-\lambda)\cdot\left[(2-\lambda)(1-\lambda)\right] = (1-\lambda)^2(2-\lambda)$$

$\lambda_1 = 1 \quad \lambda_2 = 1 \quad \lambda_3 = 2$.

$\lambda = 1$ $\quad \vec{v_1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $\quad \lambda M = 2$ $\quad GM = n - \text{rank}(A - \lambda I) = 1$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

rank = 2

block: $n - r = 3 - 2 = 1$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ rank: 2

~~block size: 1~~

$\lambda = 2$ $\quad \vec{v_3} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$ $\quad GM = 1$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & 1 \end{bmatrix} \quad R_3 \to R_3 + 2R_1 \quad \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & -8 & 1 \end{bmatrix}$$

rank = 2

block: $n - r = 3 - 2 = 1$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & -8 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & -8 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ rank = 1

~~block size = 2~~

JCF:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

JCF:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$

$A = PDP^{-1}$

$P = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$

$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{6} & \frac{1}{3} \end{bmatrix}$

$(A - \lambda I)\vec{v} = \lambda v \quad \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$

$[A - \lambda I]\vec{v} = 0$

$\det(A - \lambda I) = (3-\lambda)(-1)^{1+1}\begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + 1\cdot(-1)^{1+3}\begin{vmatrix} 0 & 3-\lambda \\ 1 & -1 \end{vmatrix}$

$= (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right] + -(3-\lambda)$

$= (3-\lambda)\left[12 - 3\lambda - 4\lambda + \lambda^2 - 2\right]$

$= (3-\lambda)(\lambda^2 - 7\lambda + 10)$

$= (3-\lambda)(\lambda-2)(\lambda-5)$

$\lambda_1 = 3 \quad \lambda_2 = 2 \quad \lambda_3 = 5$

① $\lambda = 3 \qquad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & \mathbf{3} & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$A - \lambda I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \; R_2 \leftarrow R_2 - R_1 \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 2 & -2 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$

$R_3 \leftarrow -\frac{1}{2}R_2 \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \; R_1 \leftarrow R_1 - R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$

$R_3 \leftarrow R_3 + R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 3 & 1 & -1 & 1 \end{array}\right] \; R_2 \leftarrow \frac{1}{3}R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 1 & \frac{1}{6} & -\frac{1}{6} & \frac{1}{3} \\ 0 & 0 & 3 & 1 & -1 & 1 \end{array}\right]$

$R_2 \leftarrow R_2 - R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{array}\right] \; R_3 \leftarrow \frac{1}{3}R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{array}\right]$

$P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{6} & \frac{1}{3} \end{bmatrix}$

② $\lambda = 2$

$A - \lambda I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \qquad \vec{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

③ $\lambda = 5$

$A - \lambda I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \qquad \vec{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$a\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} + b\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + c\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$$

$$\begin{cases} 2b - c = -3 \\ 2a - b + c = 8 \\ -2a + 2b = -8 \end{cases} \Rightarrow \begin{cases} a = 3 \\ b = -1 \\ c = 1 \end{cases}$$

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Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Sol: $\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ and $\begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$ are linearly independent. They span ~~the~~ $\mathbb{R}^2$ which is a subspace of $\mathbb{R}^4$.

let $u_1 = v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $\quad \|v_1\| = \sqrt{2}$

$$u_2 = v_2 - \frac{v_1 \cdot v_2}{\|v_1\|^2} \cdot v_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$$

$u_1 \cdot u_2 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix} = 0$ It is orthonormal.

Therefore, the orthonormal basis for V: $\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ and $\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

$$\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix} \quad \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}$$

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Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

proof: $(A^T)^T = A \quad -A = -(A^T)^T \qquad A = -A^T$

$a_{11} = -a_{11} \quad a_{22} = -a_{22} \quad \cdots \quad a_{nn} = -a_{nn}$ So the diagnal are zero.

$a_{11} = a_{22} = a_{33} = \cdots = a_{nn} = 0$

This is proof by example

use 3x3 square matrix as an example.

$$\begin{bmatrix} -a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} -a_{11} & -a_{21} & -a_{31} \\ -a_{12} & -a_{22} & -a_{12} \\ -a_{13} & -a_{23} & -a_{33} \end{bmatrix}$$

$\det A = a_{11}a_{12}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{21}a_{12}a_{33} - a_{11}a_{23}a_{32}$

$= a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$

Because $a_{12} = -a_{21} \quad a_{23} = -a_{32} \quad a_{31} = -a_{13}$

Then $a_{12}a_{23}a_{31} = -a_{13}a_{21}a_{32} = 0$.

We can make it to $n \times n$ square matrix.

As we know, the first one is $a_{11}a_{22}\cdots a_{nn}$ always equal to ~~equal~~ zero.

when n is odd, in the formula, the number of value whose sign before them is "+" is odd. Therefore, besides the first one, the rest of them (number of) is even.

Because ~~$a_{22} = -a_{22}$~~ $a_{ij} = -a_{ji}$, then the value ~~in it~~ whose sign in the formula is "+" add together and the sum is zero.

For the value whose sign before them in the formula is "-" all have a a_{ii} in it. Therefore all of them equals to zero.

In conclusion. $\det A = 0$.

If n is odd.

By the way. if n is even, $\det A$ may not be zero due to the "+" value in det formula is even and there would be something left.

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Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\| \leq \|u\|+\|v\|$

$|u \cdot v| \leq \|u\|\|v\|$

$(\|\vec{u}+\vec{v}\|)^2 = (\vec{u})^2 + 2\vec{u}\cdot\vec{v} + (\vec{v})^2$

$(\|\vec{u}-\vec{v}\|)^2 = (\vec{u})^2 - 2\vec{u}\cdot\vec{v} + (\vec{v})^2$

If $\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$. Then $\vec{u}\cdot\vec{v} = -\vec{u}\cdot\vec{v}$

Therefore $\vec{u}\cdot\vec{v} = 0$ $u \perp v$

$\|u+v\|^2 + \|u-v\|^2 = 2\|\vec{u}\|^2 + 2\|\vec{v}\|^2$

Therefore $\|u+v\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$

In conclusion, $\vec{u}$ and $\vec{v}$ are perpendicular to each other.